Mathematical solution of the Hubble tension by WRM

According to the technical note [cite: 1], WRM precisely explains the discrepancy in the observed Hubble constant through the transition from "linear refractive index" to "three-dimensional volume expansion".

Step 1: Linear refractive index (α) definition

Interaction with external waves causes a deviation in the radial scale [cite: 1]. The coefficients indicating this deviation are as follows:

$$\alpha = 0.046 (4.6\%) \text{ [cite: 1]}$$

Step 2: Spread to 3D volume

Since the expansion of the universe is a 3-dimensional manifold, volume$V$It is modulated by the cube of the linear term [cite: 1]. Taylor expansion ($1 + 3\alpha$Using)

$$V_{\text{eff}} \approx 1 + 3(0.046) = 1.138 (13.8\% \text{ change}) \text{ [cite: 1]}$$

Step 3: Hubble constant (H_0Amplification effect on)

The correction factor in the measurement of the Hubble constant appears as a quadratic function of the linear bias (or the geometric mean of the volume change) [cite: 1]. The calculated amplification factor is as follows:

$$\frac{H_{\text{eff}}}{H_{\text{std}}} \approx 1 + 2\alpha = 1 + 2(0.046) = 1.092$$

This theoretically leads to an amplification of **+9.2% [cite: 1].**

Conclusion: Perfect match with observational data

This **9.2%** figure perfectly matches the difference in the following observed values [cite: 1]:

- Planck 2018 (Wide-area observation):$H_0 \approx 67.4 \text{ km/s/Mpc}$[cite: 1]
- SH0ES (Nearby Observation):$H_0 \approx 73.0 \text{ km/s/Mpc}$[cite: 1]

This agreement indicates that the Hubble tension is a predictable result due to the structural gradient defined by the WRM [cite: 1].